UWIT Inventory Management System

COSC 4820 — Database Systems | Spring 2026

Dr. Amira Yousif

Abdalla Elokely | Nolan Berg | Nolan Nachbar | Thomas Hinckley | Collin Davis

PHASE 1

Project Foundation & Requirements

1. Project Domain Selection

Our team selected Inventory Management as our project domain. Specifically, we are designing a database system for the University of Wyoming IT (UWIT) Classroom Technology Services; the department responsible for deploying, tracking, and maintaining AV and technology equipment in classrooms and learning spaces across campus.

The domain was chosen because all five group members are familiar with the existing system used by UWIT, the Learning Space Manager (LSM), giving us direct access to real requirements, real pain points, and a real organizational context to work from.

Written by: Thomas Hinckley

2. Organization Overview

2.1 Organization Details

Attribute	Type	Notes
Organization		University of Wyoming IT — Classroom Technology Services
Location		University of Wyoming, Laramie, WY 82071
Current System		Learning Space Manager (LSM) — hosted at uwyo.talem3.com
System Scale		18,682 individual equipment items tracked across 1,022 campus locations

2.2 What the Organization Does

UWIT Classroom Technology Services is responsible for the installation, maintenance, and lifecycle management of AV and technology equipment in University of Wyoming classrooms, labs, conference rooms, and auditoriums. This includes projectors, displays, audio systems, touchpanels, network switches, computers, microphones, document cameras, and more.

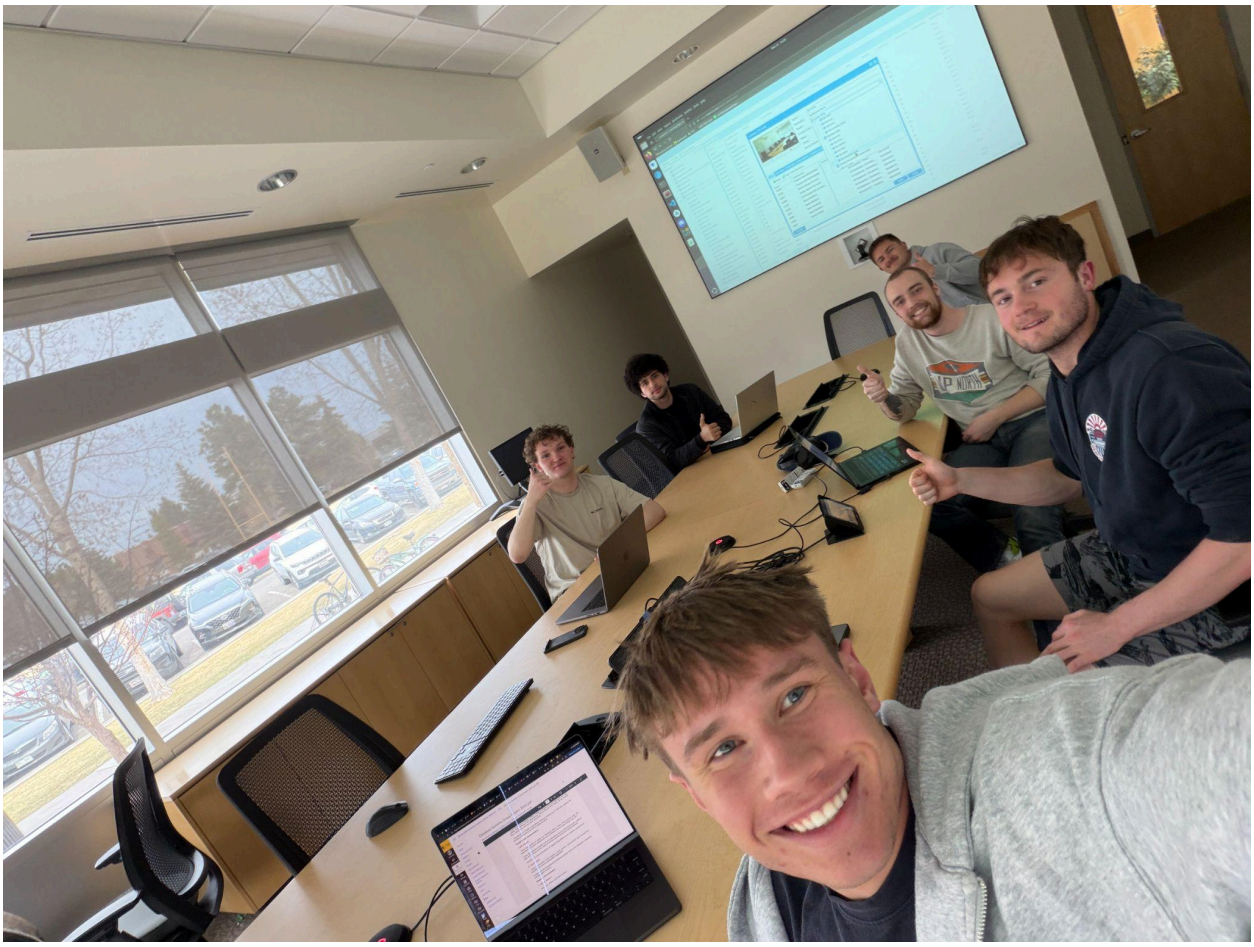
Staff use the LSM system daily to look up equipment by room, track serial numbers, log purchase dates, monitor end-of-life timelines, and identify what catalog items are deployed where across campus.

Written by: Nolan Berg

3. Organizational Visit

3.1 How We Conducted Our Visit

Our team conducted our organizational visit with UWIT Classroom Technology Services. Multiple group members are active student employees within the department, working as Classroom Technology Specialists and Inventory Specialists, giving us direct, firsthand access to the LSM system and the staff who use it daily. We observed the system in active use, consulted with department staff about current workflows, and identified pain points from our own experience managing inventory within the system.



3.2 Interview Summary

Through conversations with UWIT staff and our own experience using LSM, we gathered the following key insights:

- The system tracks equipment across three main concepts: where it is (location), what type it is (catalog), and the individual unit itself (item).
- Rooms on campus are organized by building — e.g., WU 0223 means room 0223 in the Wilbur Knight building. The current system does not formally separate building from room, making building-level queries harder.
- Every item has a catalog entry that describes the model and manufacturer. Multiple items of the same model all point to the same catalog record.
- Catalog entries are organized into hierarchical sections, such as All Equipment > Audio > Speakers. This hierarchy is currently stored as a formatted string rather than a relational structure.
- Serial numbers, purchase dates, and MAC addresses (for network equipment) are the most frequently looked-up fields on individual items.
- End-of-life dates are important for budgeting and replacement planning. Staff frequently filter for items approaching or past their EOL.
- Staff noted that the current system contains many fields that are never filled in or used, adding visual clutter without adding value.

Written by: Collin Davis

4. Requirements Gathered

4.1 Data the System Needs to Store

Based on our visit and research, the following data must be captured:

- Buildings: each campus building has a short code (e.g., AG, WU, EN) and a full name.
- Rooms/Locations: each room belongs to a building and has a type (Classroom, Lab, Conference Room, Auditorium), a capacity, and an active/inactive status.
- Catalog Sections: equipment categories organized in a parent-child hierarchy (e.g., All Equipment > Projectors).
- Catalog Entries: each unique equipment model (manufacturer + model number) belongs to a catalog section and has a life expectancy, MSRP, and description.
- Items: individual physical units, each tied to a catalog entry and a location. Key fields include serial number, purchase date, received date, end-of-life date, MAC address (for network devices), and operational status.

4.2 Organizational Goals

UWIT staff identified the following goals for an improved inventory system:

- Quickly look up all equipment in a specific room or building.
- Filter items by status (Operational, In Repair, Retired) to understand current deployment health.
- Identify items approaching or past their end-of-life date for replacement planning.
- Understand how long an item has been deployed (purchase date and received date).
- Browse the equipment catalog by category to see what models are in use across campus.

4.3 Pain Points in the Current System

The following problems were identified in the existing LSM system that our design will address:

- Building and room are combined into a single location field with no relational separation, making building-level queries require string parsing.
- Catalog sections are stored as formatted strings (e.g., 'All Equipment :: Audio :: Speakers') rather than a proper parent-child relational structure.
- Many fields on items are unused or redundant — including duplicate MAC address columns, 'Host Name', 'Reserved for', and 'UW Tag'.
- Catalog entries contain vendor logistics fields (First Ship Date, Last Ship Date, Qty Labor, Unit Labor) that are irrelevant to IT inventory management.
- No distinction between purchase date and received/deployment date, making it difficult to track how long equipment has actually been in service.

Written by: Nolan Nachbar

---

5. Entities, Relationships, and Constraints

5.1 Key Entities Identified

- Building: represents a campus building. Attributes: code, full name, and address.
- Location: represents a room within a building. Attributes: room number, type, capacity, status, and owner category.
- Catalog Section: represents an equipment category in a hierarchy. Attributes: name, parent section.
- Catalog: represents an equipment model. Attributes: manufacturer, model, section, status, life expectancy, MSRP, description.
- Item: represents a single physical unit. Attributes: code, serial number, status, purchase date, received date, EOL date, MAC address.

5.2 Key Relationships

- A Building contains many Locations (one-to-many).
- A Location holds many Items (one-to-many).
- A Catalog Section can have many child Catalog Sections (self-referencing one-to-many).
- A Catalog Section contains many Catalog entries (one-to-many).
- A Catalog entry describes many Items (one-to-many).

5.3 Key Constraints

- Every Item must reference a valid Catalog entry: an item cannot exist without knowing what type it is.
- Every Item must reference a valid Location: equipment must be assigned to a room.
- Every Location must reference a valid Building.
- Catalog Section parent_id is nullable: top-level sections have no parent.
- Item code must be unique across the system.
- Item and Catalog status fields are constrained to defined values (e.g., Operational, Retired, In Repair).

Written by: Abdalla Elokely

6. Project Scope

Our database will focus exclusively on the inventory management side of the LSM system. The following are intentionally out of scope for this project:

- Ticketing and maintenance requests
- Room assessments and LSRS scoring
- User accounts and role-based access
- Vendor and purchase order management

Limiting scope to inventory, locations, catalog, and items allows the team to build a clean, well-normalized system that directly addresses the core data management needs of UWIT without unnecessary complexity.

Written by: Whole Team

7. Team & Responsibilities

Each team member will own one table in the database, taking responsibility for its design, SQL implementation, sample data, and queries:

Group Member	Table	Domain
Abdalla Elokely	**building**	Campus buildings — codes, names, addresses
Nolan Berg	**location**	Rooms within buildings — type, capacity, status
Nolan Nachbar	**catalog_section**	Equipment category hierarchy (self-referencing)
Thomas Hinckley	**catalog**	Equipment models — manufacturer, specs, life expectancy
Collin Davis	**item**	Individual physical assets — serial numbers, dates, status

Written by: Whole Team

PHASE 2

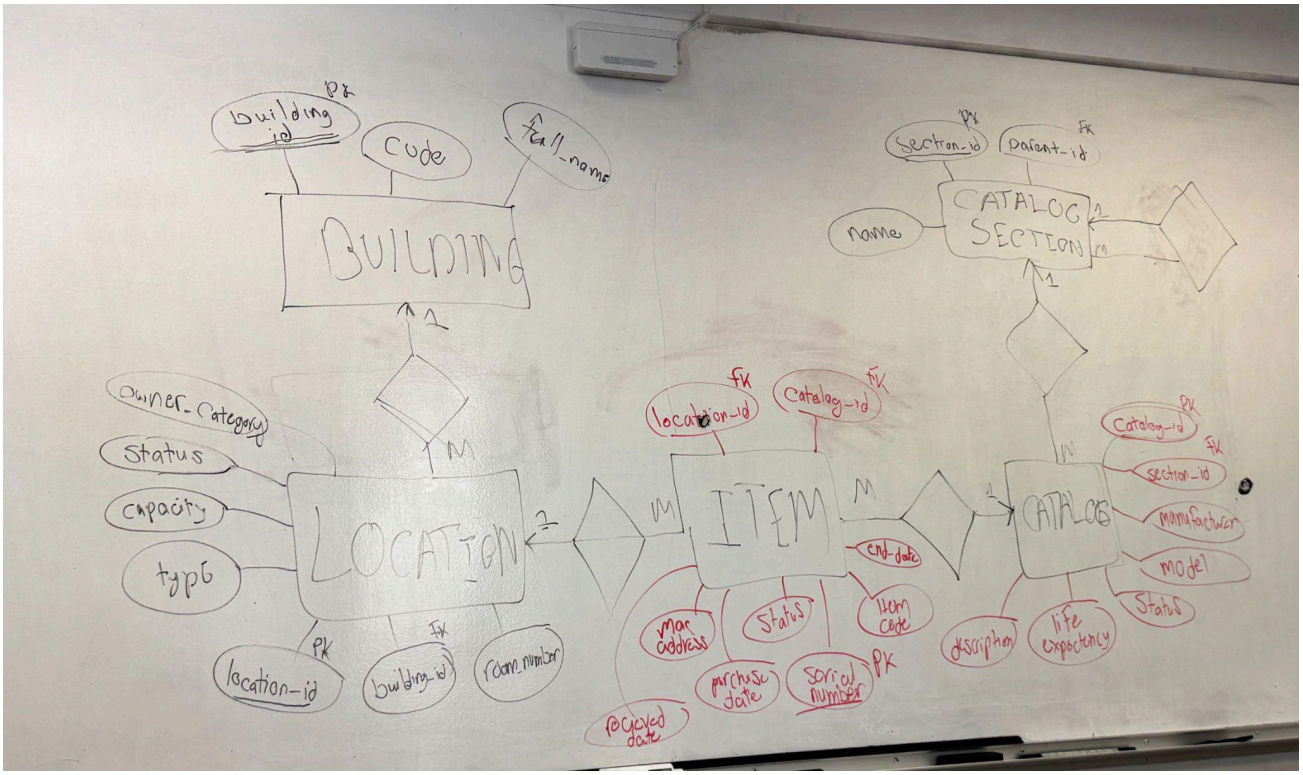
Entity & Attribute Reference

Overview

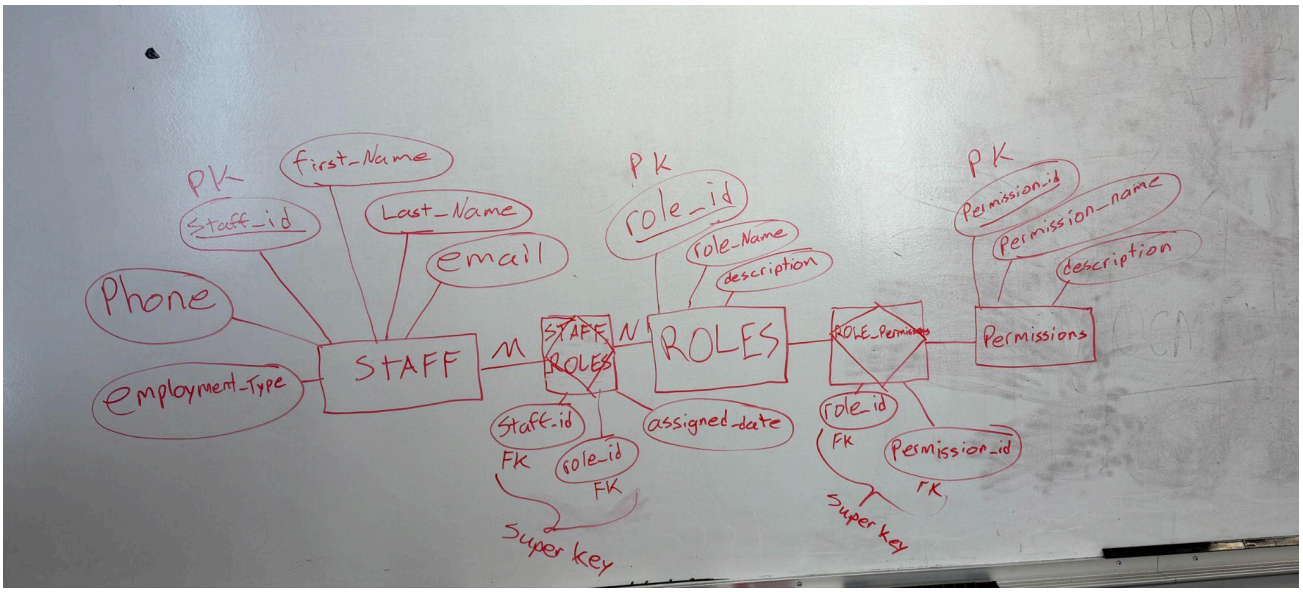
This document presents the complete entity and attribute reference for Phase 2 of the UWIT Inventory Management System database project. The schema consists of 12 tables organized into two logical groups: the core inventory layer (5 tables) modeled directly from the Learning Space Manager system, and a permissions/access control layer (7 tables) implementing role-based access control (RBAC) with staff-location assignments. All entities are normalized to 3NF.

ER Diagrams

1. Hand Drawn



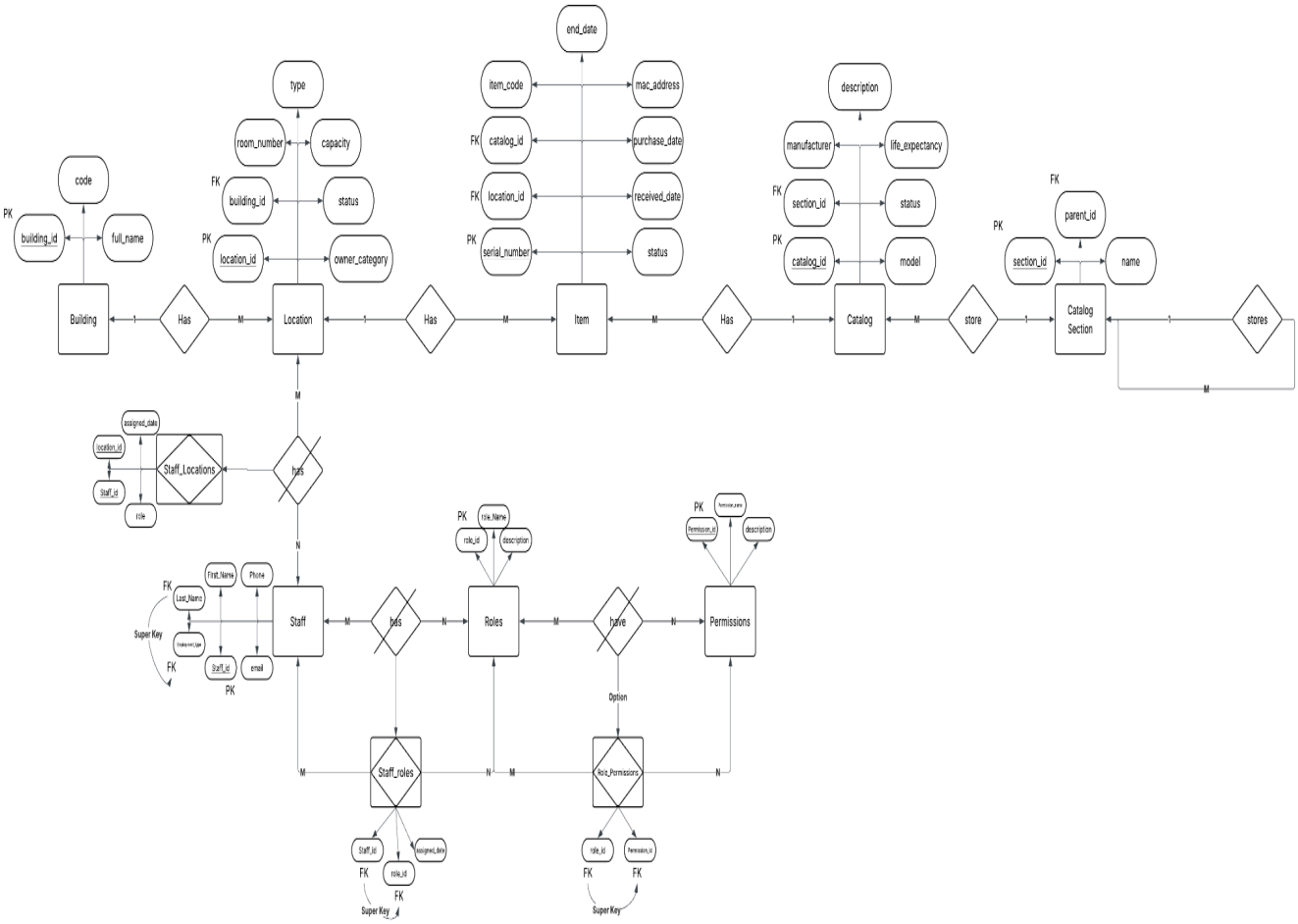
Drawn by: Abdalla Elokely



Drawn by: Nolan Nachbar

2. Digital Drawn

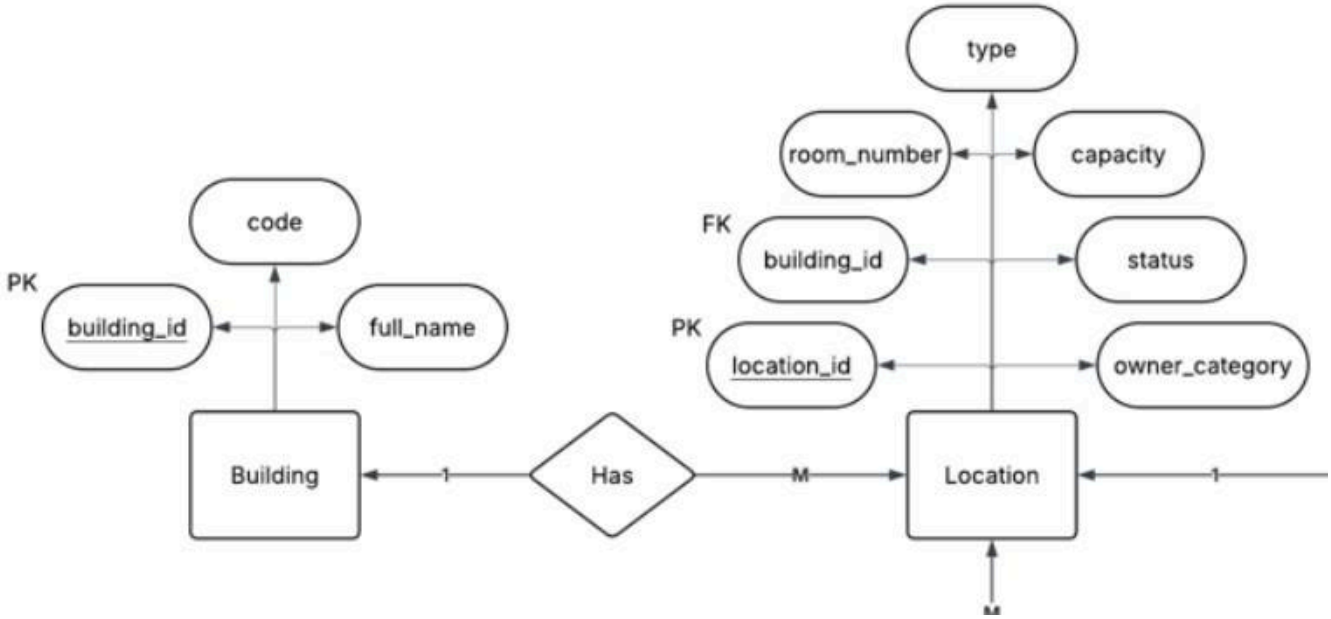
Full E-R Diagram



Build/location relationship:

- A building has one or many locations? Many.
- A location has one or many buildings? One.

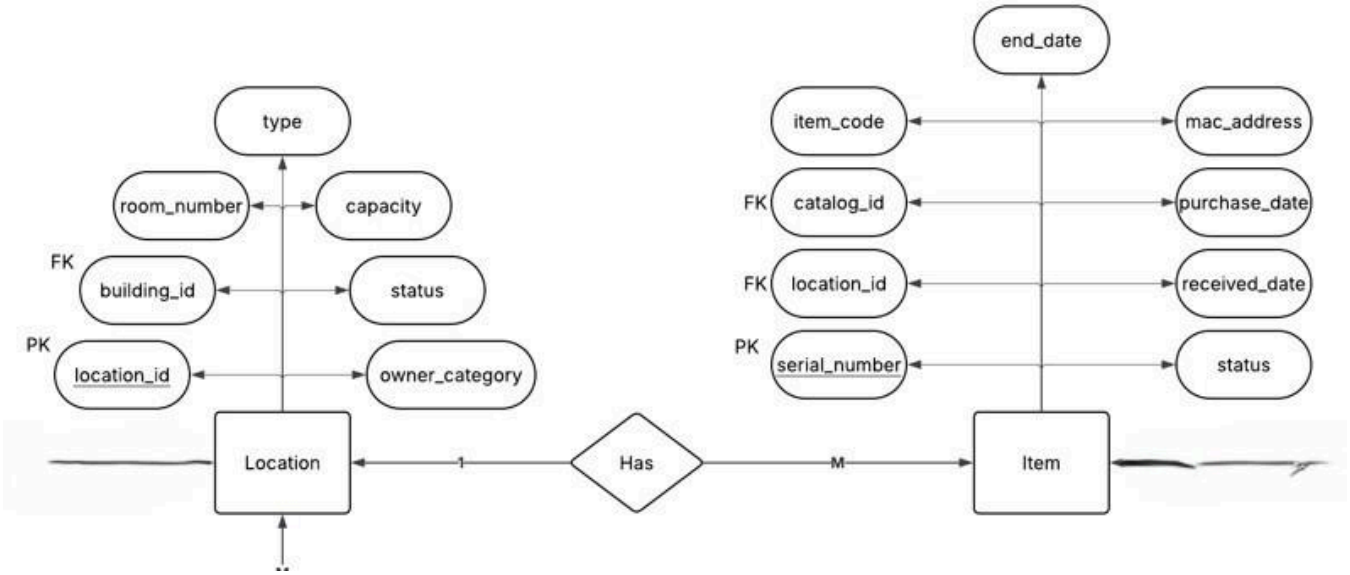
So the relationship is many to one.



Location/Item relationship:

- An item has one or many locations? One.
- A location has one or many Items? Many.

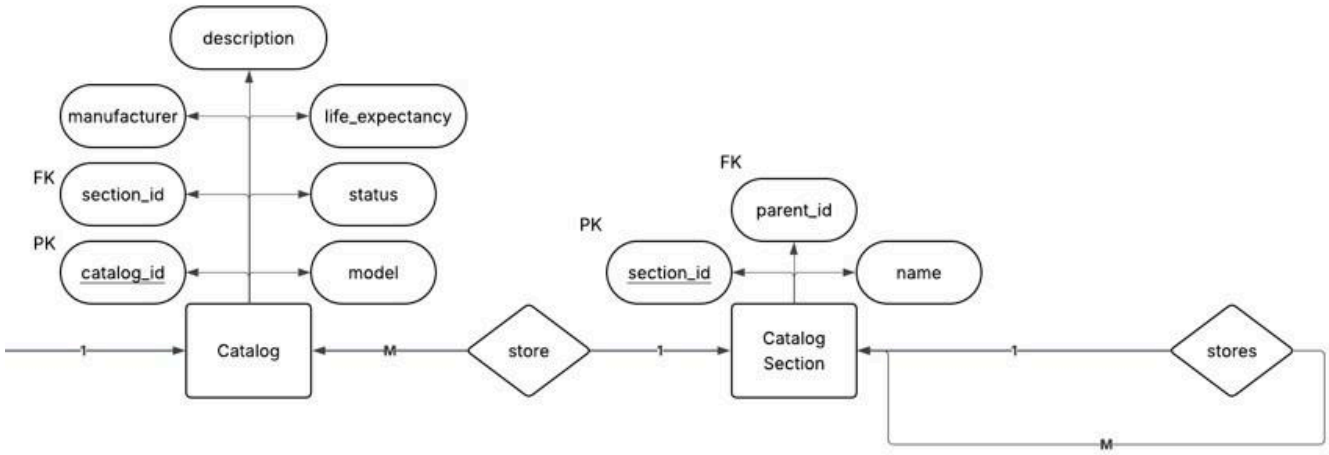
So the relationship is many to one.



Catalog Section/Catalog:

- A catalog section stores one or many catalogs? Many.
- A catalog has one or many catalog sections? One.
- a catalog section can store one or many catalog sections so it's self referencing.

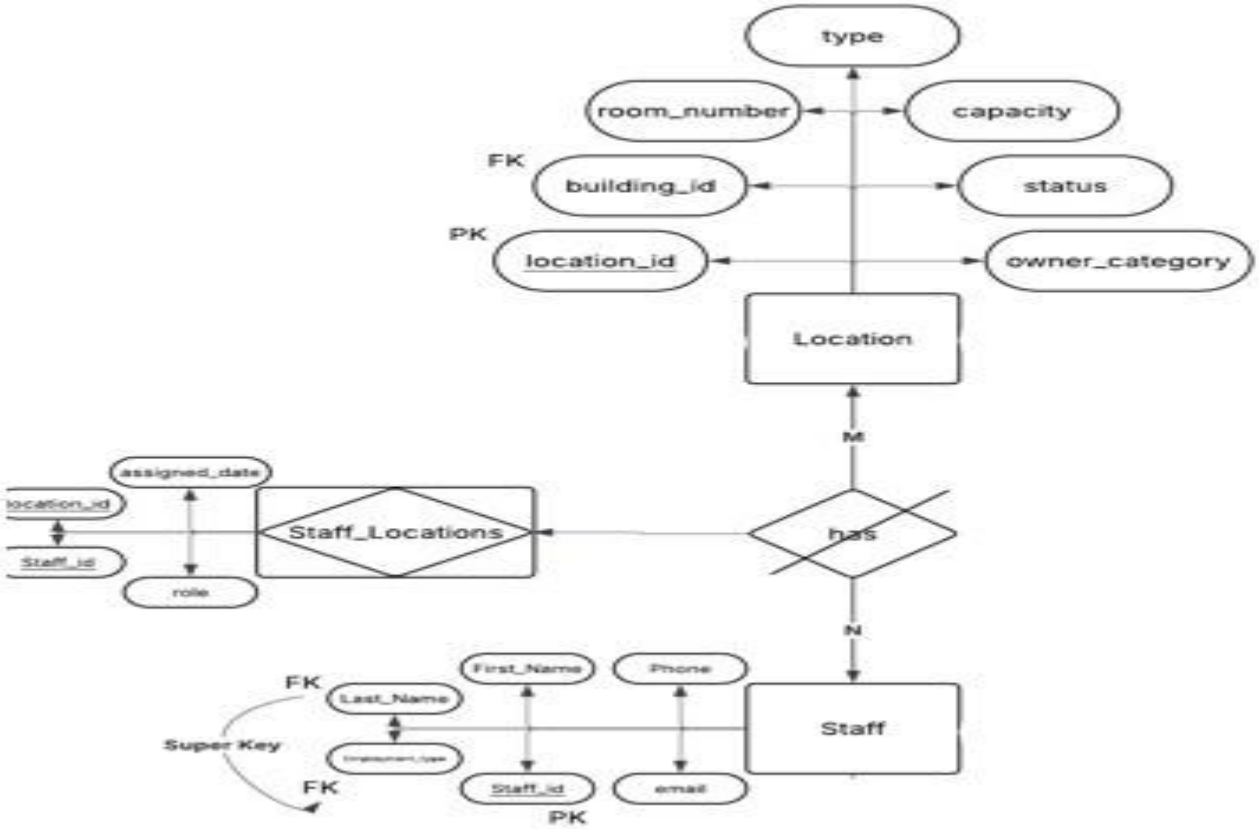
So the relationship is many to one.



Staff/Location:

- Staff has one or many locations? Many.
- A location has one or many staff members? Many.
- This creates the newborn staff_locations entity

So the relationship is many to many.

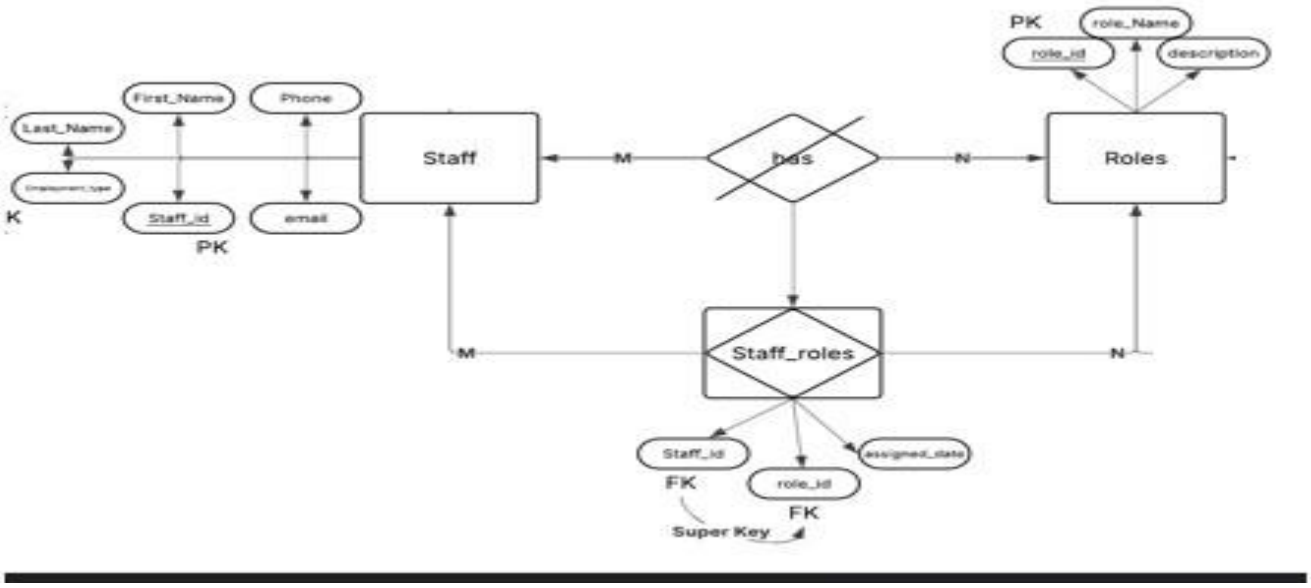


Staff/Roles:

- Staff has one or many roles? Many.
- A roleStaff has one or many staff members? Many.

- This means we have a newborn entity called staff_roles.

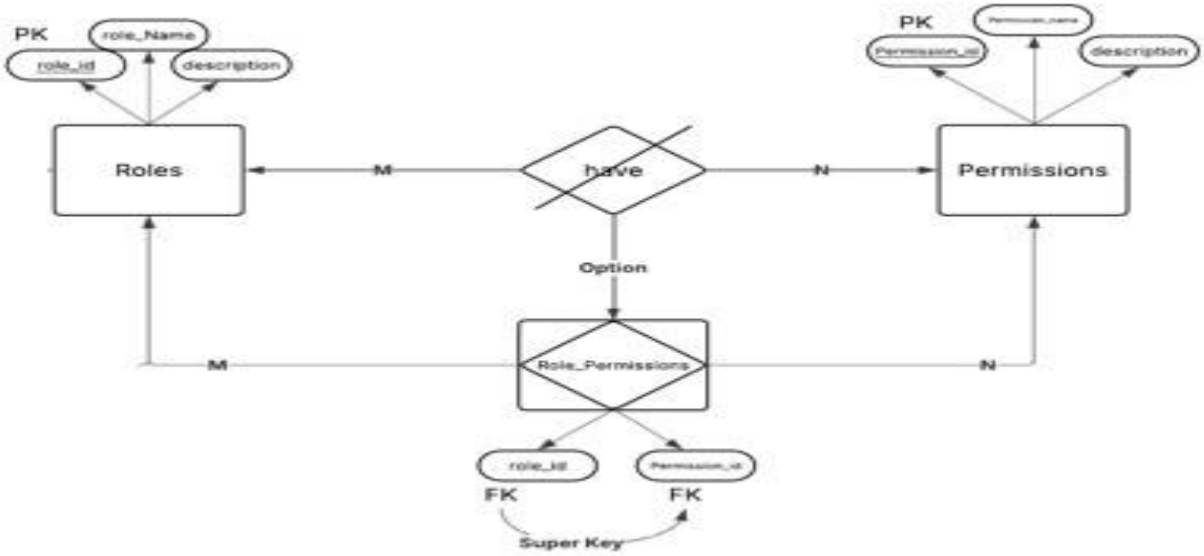
So the relationship is many to many



Permissions/Roles:

- Permission have one or many roles? Many.
- Roles have one or many permissions? Many.
- This means we have a newborn entity called Role_Permissions.

So the relationship is many to many.



Core Inventory Entities

1. Building

Owner: Nolan Berg — campus buildings, codes, and addresses

Attribute	Type	Notes
building_id	INT	Primary Key, auto-increment

code	VARCHAR(10)	Unique — e.g. "WU", "AG", "EN"
full_name	VARCHAR(100)	e.g. "Wilbur Knight Building"
address	VARCHAR(255)	Street address of building

2. Location

Owner: Nolan Berg — rooms within buildings

Attribute	Type	Notes
location_id	INT	Primary Key, auto-increment
building_id	INT	FK → building
room_number	VARCHAR(20)	e.g. "0223"
type	VARCHAR(50)	CHECK: Classroom, Lab, Conference Room, Auditorium
capacity	INT	Maximum occupancy
status	VARCHAR(20)	CHECK: Active, Inactive
owner_category	VARCHAR(100)	e.g. "Academic Affairs"

3. Catalog_Section

Owner: Nolan Nachbar — equipment category hierarchy (self-referencing)

Attribute	Type	Notes
section_id	INT	Primary Key, auto-increment
name	VARCHAR(100)	e.g. "Audio", "Speakers"
parent_id	INT	FK → catalog_section, nullable (NULL = top-level)

4. Catalog

Owner: Thomas Hinckley — equipment models, specs, and life expectancy

Attribute	Type	Notes
catalog_id	INT	Primary Key, auto-increment
section_id	INT	FK → catalog_section
manufacturer	VARCHAR(100)	e.g. "Epson", "Sony"
model	VARCHAR(100)	Model number or name
status	VARCHAR(20)	CHECK: Active, Discontinued
life_expectancy_years	INT	Used for EOL planning
description	TEXT	Nullable — optional notes

5. Item

Owner: Collin Davis — individual physical assets, serial numbers, and dates

Attribute	Type	Notes
item_id	INT	Primary Key, auto-increment
catalog_id	INT	FK → catalog
location_id	INT	FK → location
item_code	VARCHAR(50)	Unique across system
serial_number	VARCHAR(100)	Manufacturer serial number
status	VARCHAR(20)	CHECK: Operational, In Repair, Retired
purchase_date	DATE	When UWIT purchased the item
received_date	DATE	When item was deployed to room
eol_date	DATE	End-of-life, stored explicitly per unit
mac_address	VARCHAR(17)	Nullable — network devices only

Permissions & Access Control Entities

The following seven tables implement role-based access control (RBAC) with staff-location assignments. Each staff member is assigned one or more roles, and each role grants a set of permissions defining what data they can view or modify. The Staff_Location junction table explicitly tracks which staff members are responsible for which campus locations. This reflects the actual access structure within UWIT, where full-time staff hold admin-level access while student employees have scoped permissions tied to specific buildings and rooms.

6. Staff

Owner: Nolan Nachbar, System users — full-time staff and student employees

Attribute	Type	Notes
staff_id	INT	Primary Key, auto-increment
first_name	VARCHAR(50)	
last_name	VARCHAR(50)	
email	VARCHAR(100)	Unique
phone	VARCHAR(20)	Nullable
employment_type	VARCHAR(20)	CHECK: Full-Time, Student Employee

7. Role

Ownerr: Abdalla, Named access roles — e.g. Admin, Inventory Specialist, Read-Only

Attribute	Type	Notes
role_id	INT	Primary Key, auto-increment

role_name	VARCHAR(50)	e.g. "Admin", "Inventory Specialist", "Classroom Tech", "Read-Only"
description	TEXT	What this role can do

8. Permission

Owner: Thomas Hinckley, Granular action-level permissions — e.g. view_item, edit_item, retire_item

Attribute	Type	Notes
permission_id	INT	Primary Key, auto-increment
permission_name	VARCHAR(50)	e.g. "view_item", "edit_item", "retire_item", "view_catalog"
description	TEXT	Description of permission scope

9. Staff_Role (junction table)

Owner: Collin Davis, Links staff members to their assigned roles — many-to-many

Attribute	Type	Notes
staff_id	INT	FK → staff
role_id	INT	FK → role
assigned_date	DATE	Date role was assigned
(staff_id, role_id)	—	Composite Primary Key

10. Role_Permission (junction table)

Owner: Abdalla, Links roles to the permissions they grant — many-to-many

Attribute	Type	Notes
role_id	INT	FK → role
permission_id	INT	FK → permission
(role_id, permission_id)	—	Composite Primary Key

11. Staff_Location (junction table)

Owner: Team, Links staff members to the locations they are responsible for — many-to-many

Attribute	Type	Notes
staff_id	INT	FK → staff
location_id	INT	FK → location
assigned_date	DATE	Date assignment was made
role	VARCHAR(50)	Nullable — e.g. "Primary Technician", "Backup"

(staff_id, location_id)	—	Composite Primary Key

The Staff_Location table captures which technicians are assigned to which campus rooms. This supports workflows such as identifying the responsible technician for a given classroom, routing work to the correct student employee, and auditing which staff have access to specific locations. The optional role column allows a location to have both a primary and backup technician without requiring a separate table.

Relationship Summary

The table below summarizes all relationships between entities and their cardinality. Foreign key constraints enforce referential integrity throughout the schema.

Relationship	Type
Building → Location	One-to-many (via building_id)
Location → Item	One-to-many (via location_id)
Catalog_Section → Catalog_Section	Self-referencing one-to-many (via parent_id)
Catalog_Section → Catalog	One-to-many (via section_id)
Catalog → Item	One-to-many (via catalog_id)
Staff ↔ Role	Many-to-many (via Staff_Role junction table)
Role ↔ Permission	Many-to-many (via Role_Permission junction table)
Staff ↔ Location	Many-to-many (via Staff_Location junction table)

Key Constraints

- Every Item must reference a valid Catalog entry — items cannot exist without a known equipment type.
- Every Item must reference a valid Location — equipment must be assigned to a room.
- Every Location must reference a valid Building.
- Catalog_Section parent_id is nullable — top-level sections have no parent.
- item_code must be unique across the entire system.
- Item status is constrained via CHECK to: Operational, In Repair, Retired.
- Catalog status is constrained via CHECK to: Active, Discontinued.
- mac_address is nullable — only populated for network-connected devices.
- Staff_Role, Role_Permission, and Staff_Location use composite primary keys — no duplicate assignments.
- Staff_Location role column is nullable — a location assignment does not require a labeled technician role.